BRAIN
ORIGINAL ARTICLE

Circadian rhythmicity of pain sensitivity in humans

Inès Daguet,[1] Véronique Raverot,[2] Didier Bouhassira[3] and Claude Gronfier[1]

Pain intensity has been reported to fluctuate during the day in some experimental and clinical conditions, but the mechanisms underlying these fluctuations are unknown. Although the circadian timing system is known to regulate a wide range of physiological functions, its implication in pain regulation is largely unknown.
Using highly controlled laboratory constant-routine conditions, we show that pain sensitivity is rhythmic over the 24 h and strongly controlled by the endogenous circadian timing system. We found that the circadian component of pain sensitivity can be modelled with a sinusoidal function, with a maximum in the middle of the night and a minimum in the afternoon. We also found a weak homeostatic control of pain sensitivity, with a linear increase over the 34 h of prolonged wakefulness, which slowly builds up with sleep pressure. Using mathematical modelling, we describe that the circadian system accounts for ∼80% of the full magnitude of pain sensitivity over the 24 h, and that sleep-related processes account for only ∼20%.
Overall, our data reveal the neurobiological mechanisms involved in driving the rhythmicity of pain perception in humans. We show that pain sensitivity is controlled by two superimposed processes: a strong circadian component and a modest homeostatic sleep-related component. Our findings highlight the need to consider time of day in pain assessment, and indicate that personalized circadian medicine may be a promising approach to pain management.

1 Lyon Neuroscience Research Center (CRNL), Neurocampus, Waking Team, Inserm UMRS 1028, CNRS UMR 5292, Université Claude Bernard Lyon 1, Université de Lyon, 69000, Lyon, France
2 Centre de Biologie et de Pathologie Est, Groupement Hospitalier Est, Hospices Civils de Lyon, 69677, Bron cedex, France
3 Inserm U987, APHP, UVSQ, Paris-Saclay Université, CHU Ambroise Paré, 92100, Boulogne-Billancourt, France

Correspondence to: Claude Gronfier
Inserm U1028, Lyon Neuroscience Research Center (CRNL)
95, Boulevard Pinel, 69500 Bron, France
E-mail: claude.gronfier@inserm.fr

Keywords: pain; circadian; rhythm; sleep; homeostasis

Abbreviations: CT = circadian time; CR = constant routine; DLMOn = dim light melatonin onset; ET_{50} = half-maximal effective temperature; VAS = visual analogue scale

Introduction

Pain intensity has been reported to fluctuate during the day in a number of clinical conditions.[1] The cyclic nature of some headaches[2,3] and the diurnal variation of pain related to osteoarthritis are classical clinical observations.[4,5] The mechanisms underlying these fluctuations, however, are unknown. In particular, it remains unclear whether such daily variations are related to the internal circadian clock, or to behavioural or environmental factors, such as the sleep–wake cycle or the rest–activity cycle.

Pain has two main interconnected components: a sensory-discriminative component (location, quality, duration, intensity, etc.) and an emotional component (unpleasantness, anxiety, motivation, etc.).[6] This multidimensional nociceptive response involves the activation of numerous subcortical and cortical regions of the brain (e.g. somatosensory cortices, insula,

Received December 20, 2021. Revised March 23, 2022. Accepted April 07, 2022. Advance access publication July 11, 2022
© The Author(s) 2022. Published by Oxford University Press on behalf of the Guarantors of Brain. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

thalamus, prefrontal cortex), often referred to as the 'pain matrix'.[7] These structures are known to be regulated by the sleep/wake cycle or the circadian clock,[8–10] but it remains unclear whether pain sensitivity is rhythmic and how it is regulated.

The circadian timekeeping system plays a key role in physiology by regulating the rhythmicity of numerous functions, from gene expression to cortical activity and behavioural functions.[8,9,11–14] It is, therefore, also likely to be involved in pain perception. The surprising lack of knowledge about the rhythmicity of pain sensitivity may result from the impact of timing on pain perception rarely having been taken into account,[1] and the use of inappropriate protocols for the exploration of pain rhythmicity from a neurobiological and mechanistic point of view. The experimental studies performed to date to investigate pain sensitivity changes during the day in healthy individuals have reported conflicting results.[15–20] Indeed, one half of these articles reported no statistically significant diurnal change in pain perception[18–20] and the other half[15–17] reported a time-of-day effect on pain sensitivity but with a large variation in the timing of maximal pain (between 18:00 and 2:00). In both experimental and clinical studies, the limited number of measurements and their timing (mostly during daytime) made it impossible to demonstrate unequivocally the existence of a 24-h rhythmicity in pain sensation. It is also impossible to determine the origin of any rhythmicity in pain from these studies, because neither of the two types of highly controlled laboratory protocols [constant routine (CR) and forced-desynchrony paradigm] capable of separating endogenous and exogenous rhythms were used.[21,22] Endogenous rhythms are controlled by the central circadian clock located in the suprachiasmatic nuclei of the hypothalamus, and exogenous rhythms depend on behavioural or environmental changes, such as the sleep–wake cycle, the dark–light cycle or the rest–activity cycle. In real-life conditions, endogenous and exogenous influences are expressed simultaneously, making it impossible to attribute rhythmicity to one or the other. In this study, we aimed to determine whether thermal pain sensitivity displays rhythmicity over the 24-h day, and to assess the precise contribution of the circadian clock and sleep-related processes, by systematically assessing pain sensation and gold-standard markers of circadian rhythmicity in highly controlled CR conditions.

Materials and methods

Participants

Twelve healthy males (20–29 years old, mean age = 22.7 ± 3.3 years; BMI = 21.8 ± 3.1 kg/m^2; height = 1.78 ± 0.07 m; weight = 69.3 ± 12.7 kg) were included in this study. Neurological, psychiatric and sleep disorders were excluded by clinical examination and psychological questionnaires (Pittsburg Sleep Quality Index Questionnaire and Beck Depression Inventory).[23,24] Participants had an intermediate chronotype (mean Horne and Ostberg score = 53.5 ± 5.7),[25] and had not done any shift work or experienced transmeridian travel during the previous 3 months. Participants had normal visual acuity (Landolt ring test and Monoyer scale), contrast vision (Functional Acuity Contrast Test) and colour vision (Farnworth D-15 and Ishihara colour test). All experimental procedures were carried out in accordance with the Declaration of Helsinki between March and June 2017. The study was approved by the local research ethics committee (CPP Lyon Sud-Est II) and participants provided written informed consent for participation.

Study design

Participants were asked to maintain a regular sleep/wake schedule (bedtimes and waketimes within ± 30 min of self-targeted times) for an average of 3 weeks before admission to the laboratory, with verification by wrist activity and light exposure recordings (ActTrust, Condor Instruments). Habitual bedtimes were determined on the basis of sleep times averaged over the 7 days preceding the laboratory segment of the protocol. Average bedtime was 23:45 and average waketime was 8:00. Subjects were then admitted to the laboratory for a 56-h experimental protocol (Fig. 1), in which they were kept in an environment free from external time cues (clocks, television, smartphones, internet, visitors, sunlight, etc.). Subjects maintained contact with staff members specifically trained to avoid communicating time-of-day information or the nature of the experimental conditions to the subjects. Participants arrived at ~10:00 on the first day. They were allowed to familiarize themselves with the laboratory environment, low light levels (<0.5 lx), equipment and measurements. Lunch and dinner were served at ~12:30 and ~19:00. A series of measurements were then performed until bedtime (participant's habitual bedtime), and an 8-h sleep episode was scheduled (constant darkness; recumbent position). This was followed by a 34-h CR protocol beginning at the participant's usual waketime on Day 2, and ending on Day 3 (18:00 on average).

Constant-routine protocol

A CR paradigm was used to reveal the endogenous circadian rhythmicity of various parameters. The CR was conducted under constant environmental conditions to eliminate, or distribute across the circadian cycle, the physiological responses evoked by environmental or behavioural stimuli (i.e. sleeping, eating, changes in posture, light intensity variations).[21,26] In practical terms, participants were asked to remain awake for 34 h (starting at their habitual waketime), with minimal physical activity, while lying in a semi-recumbent (45°) posture in bed. This posture was also maintained for the collection of urine samples and bowel movements. Room temperature [mean = 23°C ± 0.6 (SD)] and ambient very dim halogen light levels were kept constant. Light intensity was homogeneous in the room (<0.5 lx at the participant's eye level in all directions of gaze). Participants were given small equicaloric snacks and fluids at hourly intervals to maintain an equal nutritional caloric intake and stable hydration over the circadian cycle. Caloric requirements were calculated on the basis of basal metabolic rate determined with the Wilmore nomogram and were adjusted upward by a 7% activity factor.[27,28] Fluid intake was calculated for each subject, to account for the sedentary nature of the CR,[28] and consisted only of water (no caffeinated beverages). A member of the study staff remained in the room with the participant at all times during the CR, to monitor wakefulness and to ensure compliance with the study procedures. In between measurements, participants were free to do any activity that respected the constraints of the CR protocol. The predominant activities were reading, writing, drawing, chatting and listening to music or podcasts.

Heat and pain evaluation

Thermal stimuli were applied to the forearm with a Peltier-type thermode (30 × 35 mm) connected to a thermotest device (Somedic AB). All heat/pain sensitivity measures (heat detection threshold, heat pain threshold and half-maximal effective temperature (visual analogue scale, VAS) pain intensity evaluations to 2-s heat stimuli at 42°C, 44°C and 46°C) were conducted every 2 h.

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

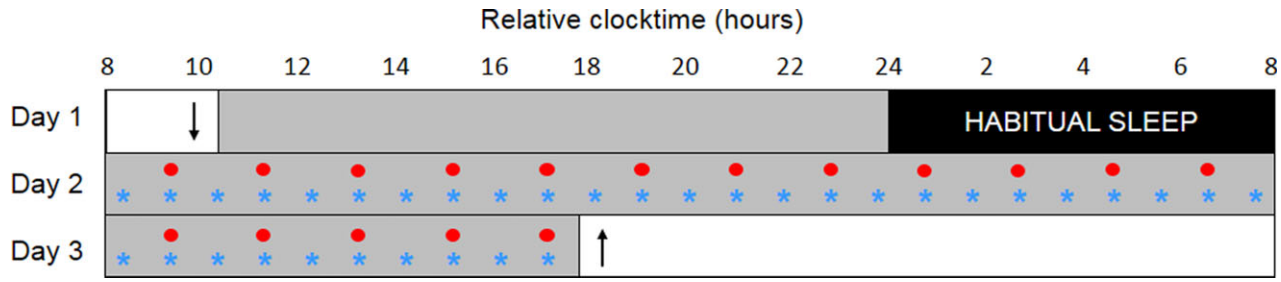

Figure 1 Overview of the experimental protocol. After a day of habituation (Day 1) and an 8-h sleep episode, participants were subjected to a 34-h CR (Days 2 and 3). Melatonin levels were assessed hourly (asterisks); heat/pain sensitivity (thresholds and intensity), temperature, and heart rate were evaluated every 2 h (filled circles). Participants arrived at ~10:00 on Day 1 (down arrow) and left the laboratory at ~18:00 on Day 3 (up arrow). Grey rectangles represent wakefulness in dim light (~0.5 lx) and black rectangles represent scheduled sleep in darkness.

Heat detection and pain thresholds were determined according to the method of limits (mean of three measurements). Thermode temperature was gradually increased from a baseline temperature of 32°C, at a rate of 1°C/s, and participants were asked to stop the increase in temperature when they started to feel a warm sensation (detection threshold) or a pain sensation (pain threshold). At this point, the temperature returned to baseline at a rate of 1°C/s. A minimum interval of 20 s was respected between each threshold measurement. If participants had not pressed the button by the time the maximum temperature (50°C) was reached, the stimulation was stopped and the maximum temperature was recorded as the threshold value. In this study, all participants (and at all time points) pressed the button to indicate their detection or pain threshold; the maximum temperature of 50°C was never reached.

The pain induced by graded thermal stimuli was assessed with a 100-mm VAS. All participants also received stimulation with three pseudorandomized heat stimuli (42°C, 44°C and 46°C). For each stimulus, participants were asked to rate the intensity of the pain on a VAS, extending from 'no pain' to 'maximal imaginable pain'. For each stimulation, the thermode temperature gradually rose from baseline temperature (32°C) at a rate of 1°C/s. Once the target temperature was reached, it was maintained for 2 s and the temperature then returned to baseline. Stimuli were separated by an interval of at least 45 s. Pain sensitization was prevented by applying the thermode to adjacent regions of the forearm, never using the same site for consecutive stimuli.

For more precise assessments of pain sensitivity than could be achieved with the responses to arbitrary temperatures, intensity response curves were calculated (Fig. 3). This is a better approach to the assessment of sensitivity, as it can be used to determine the half-maximal effective temperature, or ET_{50}, corresponding to the stimulation temperature required to induce 50% of the maximal response (pain intensity of 5/10).

The data were modelled with a sigmoidal function:

$$f(temperature) = \max + \frac{\min - max}{1 + \left(\frac{temperature}{ET_{50}}\right)^{hillslope}} \quad (1)$$

As the VAS is a bounded scale, minimum (min) and maximum (max) pain scores were set at 0 and 10, respectively. Hillslope (slope of the curve) and ET_{50} were left free. Temperatures were expressed on a logarithm scale. The statistical power of the modelling approach was increased by calculating sigmoidal fits over 4-h time epochs, corresponding to two evaluations of pain sensitivity for each of the three stimuli (42°C, 44°C and 46°C), providing six points on the regression curve (Supplementary Fig. 1). The ET_{50} values were extracted from each of the nine sigmoidal regressions (see previous equation; Fig. 3A) and plotted over time (Fig. 3B).

Body temperature

Core body temperature was measured every 2 h, with an ear thermometer (Braun Thermoscan Pro 6000, Welch Allyn). Body temperature was measured within 2–3 s, with a precision of 0.2°C.

ECG

An ECG was continuously recorded with two adhesive skin electrodes (BlueSensor N, Ambu) positioned on the sternum and the lateral thorax (Fontaine bipolar precordial leads). The signal was recorded at 256 Hz, with a Vitaport 4 digital recorder (Temec Instruments), to assess autonomic nervous system activity. Heart rate and heart-rate variability were analysed on the basis of the bipolar ECG signal. R-wave peak detection was performed over 10-s windows during a 4.5-min baseline resting episode. For interval analysis, data were resampled at a rate of 10 Hz. Root mean square of successive differences (RMSDD) was determined to estimate the vagally (parasympathetic) mediated changes reflected in heart-rate variability.[29,30] It was not possible to obtain ECG data for the first participant, for technical reasons, so ECG analysis was performed for 11 participants.

Melatonin

Saliva was collected hourly, with cotton swabs placed directly in the mouth of the participant (Salivettes, Sarstedt). Samples were stored at −20°C until centrifugation and assay. Melatonin levels were measured with an in-house radioimmunoassay ^{125}I (RIA). This assay was based on a competition technique. The radioactive signal, reflecting the amount of ^{125}I-labelled melatonin, was therefore inversely proportional to the concentration of melatonin in the sample. The sensitivity of the assay was 1.5 pg/ml. The inter-assay coefficients of variation for high (18.5 pg/ml) and low (10 pg/ml) melatonin-concentration controls were 19 and 22%, respectively, and the mean intra-assay coefficient of variation was <10%. We determined the circadian melatonin profile of each participant over a 24-h day, by applying a three-harmonic regression individually to the raw data collected during the CR (Days 2 and 3).[31,32]

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

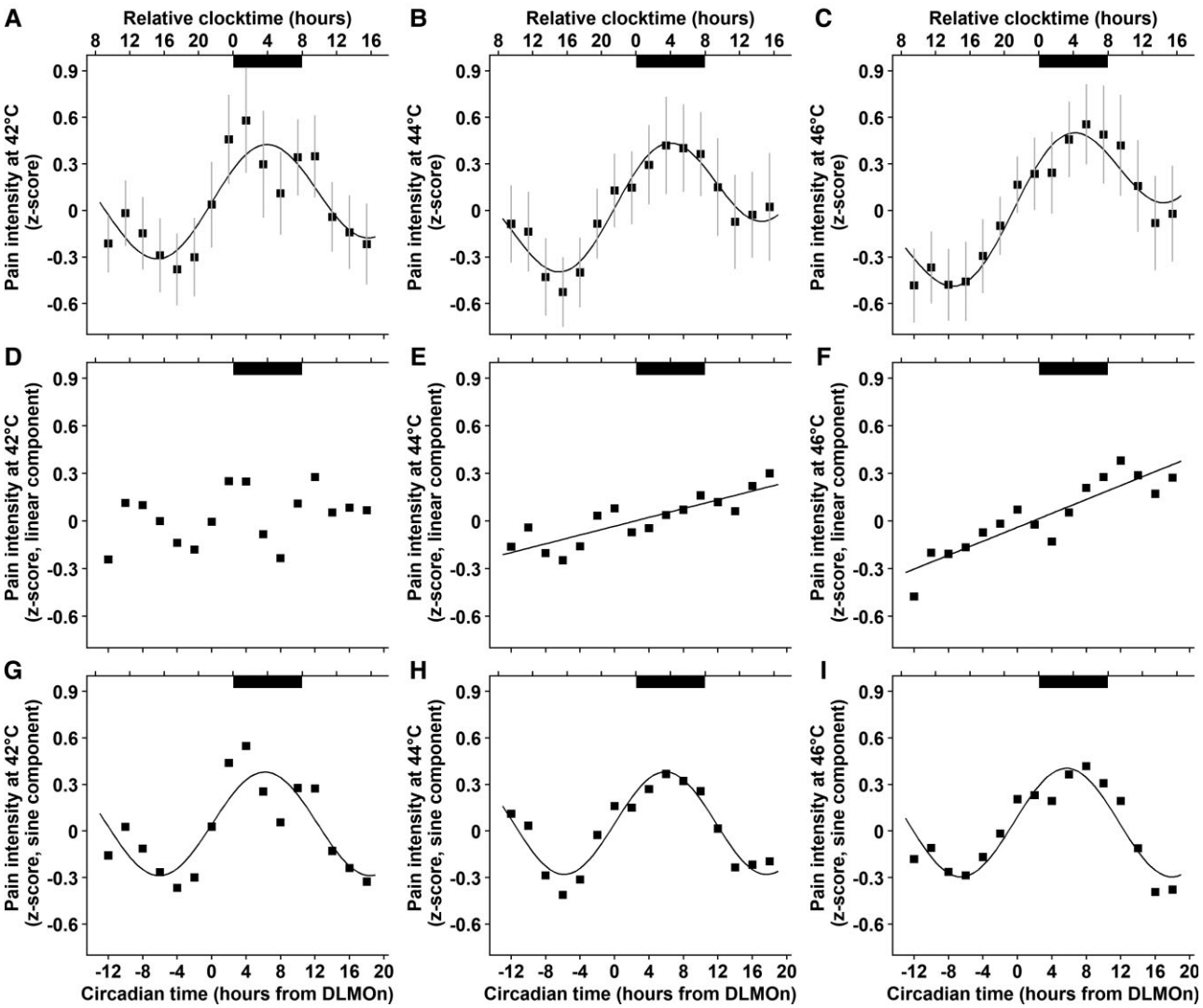

Figure 2 Mean pain intensities in response to 2-s heat stimuli at 42°C, 44°C and 46°C are rhythmic across the 34-h constant routine protocol ($n = 12$). Dark bars correspond to the average timing of habitual sleep episodes (biological night). Circadian time 0 (CT 0) corresponds to dim light melatonin onset (DLMOn, mean ≃ 21:30). (**A–C**) Combined models (sum of linear and sinusoidal components) applied to normalized data (mean ± SEM) for stimuli at 42°C (**A**, $R^2 = 0.72$), 44°C (**B**, $R^2 = 0.92$) and 46°C (**C**, $R^2 = 0.92$). (**D–F**) Linear components for stimuli at 42°C (**D**, $R^2 = 0.10$; $P = 0.23$), 44°C (**E**, $R^2 = 0.73$; $P < 0.0001$) and 46°C (**F**, $R^2 = 0.81$; $P < 0.00001$). Pain sensitivity increases with time spent awake for stimuli at 44°C and 46°C. (**G–I**). Sinusoidal components for stimuli at 42°C (**G**, $R^2 = 0.70$), 44°C (**H**, $R^2 = 0.90$) and 46°C (**I**, $R^2 = 0.86$). Pain sensitivity follows a circadian rhythm, with maximal pain at 3:30 (42°C and 44°C) or 3:00 (46°C).

The model equation was:

$$f(time) = mesor + amplitude1 \times \cos\left(2\pi \times \frac{time}{tau} + phase1\right) + amplitude2 \times \cos\left(2\pi \times \frac{time}{\frac{tau}{2}} + phase2\right) + amplitude3 \times \cos\left(2\pi \times \frac{time}{\frac{tau}{3}} + phase3\right) \quad (2)$$

In the model, Tau (the circadian period) was constrained between 23.5 and 24.5 h; mesor, amplitudes (1 to 3) and phases (1 to 3) were set free.

The dim light melatonin onset (DLMOn), corresponding to the circadian phase, was calculated for each participant. DLMOn was defined as the time at which the ascending phase of the melatonin profile crossed the 25% threshold of the peak-to-trough amplitude of the fitted curve. Because of technical problems with some saliva samples, the full 24-h melatonin profile could not be obtained for two participants. For one of these participants, DLMOn was calculated on the basis of melatonin levels during the habituation day (Day 1), rather than during the CR, for which we could not determine melatonin concentrations. For the second participant, in the absence of melatonin-concentration data (flat profile below the limit of quantification of the assay), DLMOn was estimated from the mean phase angle calculated between habitual bedtime and DLMOn (calculated from data published by Gronfier *et al.*, 2004).[31] The average DLMOn was 21:36 ± 01:08 in our group of subjects.

Statistics

Outliers were identified on the basis of individual normalized data (z-scores) and were excluded from subsequent analyses

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

(outlier.test, R, Version 3.6.1 - 2019-07-05, R Foundation for Statistical Computing, Vienna, Austria). Over the 204 values obtained per variable (17 samples obtained in 12 participants), a maximum of one outlier value was detected and removed per variable. We reduced inter-individual variability, by normalizing all subjective data by calculating individual z-scores and smoothing them with a moving average (calculated on three points). The endogenous circadian phase was taken into account for each participant, by aligning the data with the onset of melatonin secretion (DLMOn). As DLMOn occurred at different times in different participants, individual melatonin onset values were set to 0 [DLMOn = circadian time (CT) 0], and all measurement times were expressed relative to melatonin onset. We modelled the effects of time on the responses observed during the 34-h CR, using an additive model including a linear component (homeostatic, process S) and a sinusoidal component (circadian, process C). The equation of the combined model was:

$$f(time) = y0 + a \times time + mesor + amplitude \times \cos\left(2\pi \times \frac{time}{tau} + phase\right) \quad (3)$$

Tau (circadian period) was constrained between 23.5 and 24.5 h,[33,34] whereas all other parameters were left free. Once the parameters of the combined model had been defined, process S and process C were modelled separately. The homeostatic component (process S) was regressed against the linear component of the model:

$$f(time) = y0 + a \times time \quad (4)$$

The circadian rhythmicity (process C) of the data was regressed against the sinusoidal component of the model:

$$f(time) = mesor + amplitude \times \cos\left(2\pi \times \frac{time}{tau} + phase\right). \quad (5)$$

Statistics were calculated with R (v.3.6.1: 2019-07-05, R Foundation for Statistical Computing, Vienna, Austria). Results were considered significant if $P < 0.05$. Results are expressed as mean ± SD when reporting participant demographics/anthropometrics in the 'Materials and methods' sections and as mean ± SEM for data presented in the results section (unless stated otherwise).

Data availability

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Results

Pain rhythmicity is regulated by homeostatic and circadian processes

Twelve healthy males aged 22.7 ± 3.3 years (mean ± SD) participated in a 56-h experimental protocol (Fig. 1) including a 34-h highly controlled CR designed to unmask endogenous rhythmicity (enforced wakefulness, constant posture, low physical and cognitive activity, constant dim light, equicaloric snacks every hour).[21] We assessed the effect of the time of day on pain sensitivity, by measuring heat pain every 2 h during the 34 h of CR. In accordance with the current view that two main processes regulate sleep,[35] and in agreement with studies showing that subjective sleepiness,[13,36] physiological functions (such as executive functions[37]), and cortical brain responses (measured by EEG[14] and functional MRI[8]) are influenced by both sleep pressure and the circadian timing system, we then modelled the effect of time on pain with an additive mathematical model including a linear component (sleep-related homeostatic drive, process S) and a sinusoidal component (circadian drive, process C).[13]

Pain sensitivity increases with sleep debt

We probed subjective pain (VAS ratings) in response to 2-s heat stimuli (42°C, 44°C and 46°C) every 2 h over the entire 34-h CR (Fig. 2A–C and Supplementary Fig. 2; all $R^2 > 0.72$). A linear component was observed for the stimuli at 44°C and 46°C (Fig. 2E and F; all $P < 0.0001$; all $R^2 > 0.73$), but not for the less painful stimuli at 42°C (Fig. 2D; $P = 0.23$; $R^2 = 0.10$). Our results thus confirm the known relationship between sleep deprivation and greater pain sensitivity,[38–40] but also suggest that this relationship may not apply to low levels of pain. As the participants were in a constant state of wakefulness during the CR, the linear component of our model translates the effect of sleep debt and reflects homeostatic sleep pressure. The slope of the linear regression line increased with stimulation temperature between 44°C and 46°C (Fig. 2E and F), so the largest changes in amplitude were observed for stimuli at 46°C, which caused a change in pain level of 1/10 on the VAS. As pain responses were measured at three arbitrary temperatures, we then used a modelling approach (classically used in pharmacology and photobiology[41,42] to extract an overall pain sensitivity value, Fig. 3). The mathematically modelled sigmoidal intensity response curve (based on the combined results obtained at 42°C, 44°C and 46°C) yielded sensitivity values (ET_{50}) that confirmed the results reported previously; a linear increase in sensitivity to pain with time awake (lower ET_{50} values) (Fig. 3C; $R^2 = 0.81$; $P < 0.01$).

Pain sensitivity is driven by the circadian timing system, with maximal pain experienced in the middle of the night

Subjective measurements of pain in response to 2-s thermal stimuli (42°C, 44°C and 46°C) revealed that pain sensitivity was influenced not only by sleep pressure, but also by the circadian timing system (Fig. 2A–C; all $R^2 > 0.72$). Indeed, independently of the effect of sleep pressure, a sinusoidal component in our model strongly accounted for changes in pain sensitivity across the 34 h of CR, with a pain sensitivity peak between 3:00 and 4:30 for both the responses to graded stimuli (Fig. 2G–I) and heat pain thresholds obtained using stimulation ramps (Supplementary Fig. 3C). These results were confirmed by the modelling of a sigmoidal intensity response curve, which also showed a strong circadian rhythmicity of pain sensitivity (Fig. 3D; $R^2 = 0.93$) and a pain peak in the middle of the night (at 4:30). Interestingly, the lack of circadian rhythmicity for warm non-painful stimuli (Supplementary Fig. 4C; $R^2 = 0.13$) suggests that the rhythmicity of pain sensitivity is specific to pain and is not related to a general rhythmicity of thermal sensitivity.

Changes in pain sensitivity over the 24-h day are primarily induced by the circadian timing system rather than by sleep pressure

We investigated the relative contributions of sleep pressure and circadian drives to pain sensitivity, by calculating the mean changes in both these components and expressing them relatively to the total amplitude over 24 h (Fig. 2 and Supplementary Fig. 5). Using the 24 h profile at 44°C, we find that the amplitude of pain sensitivity is

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

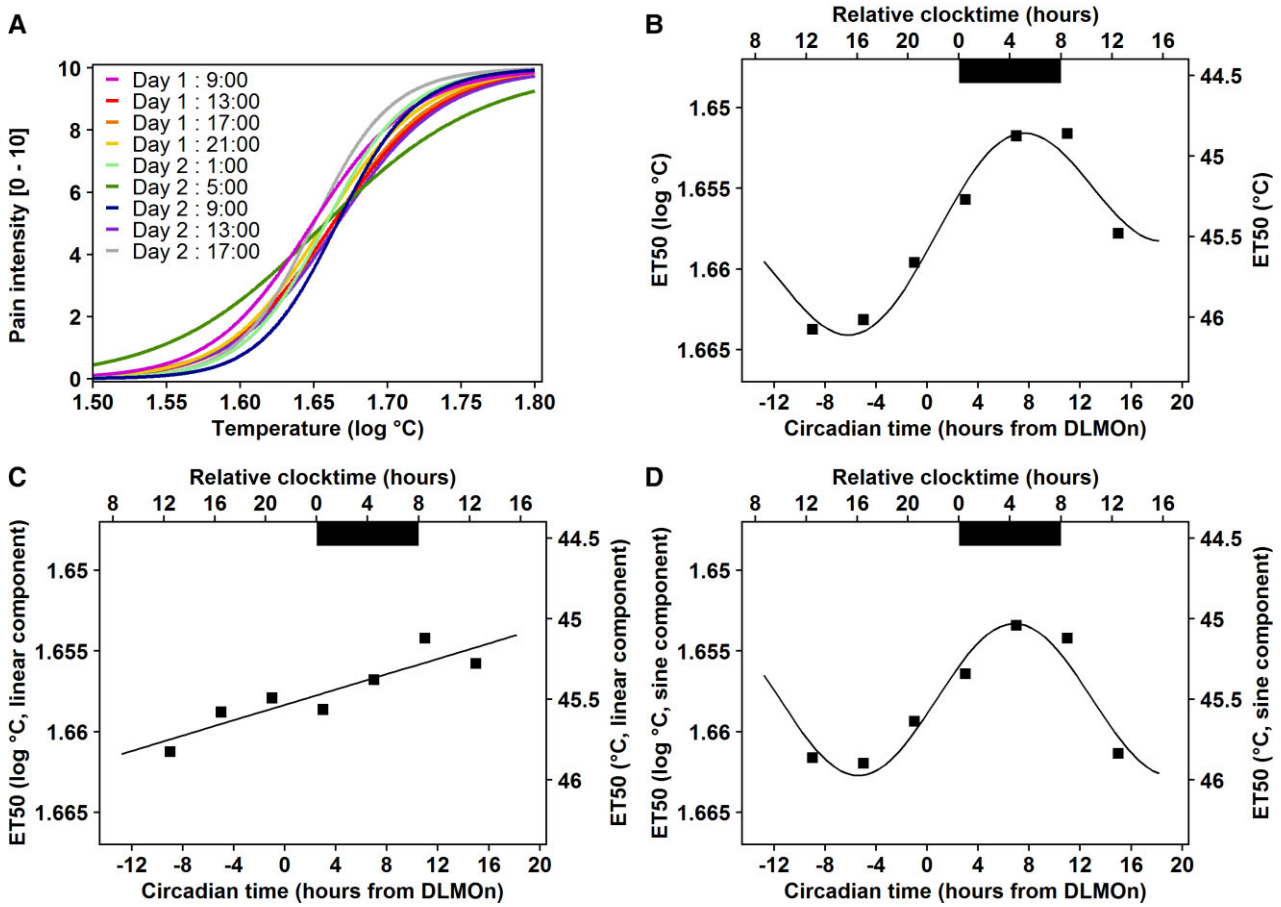

Figure 3 Mean pain sensitivity (ET_{50}) is rhythmic across the 34-h constant routine protocol ($n = 12$). (**A**) Intensity response curves calculated on six measures obtained at 42°C, 44°C and 46°C over two consecutive 2-h segments (nine curves; all R^2 between 0.68 and 0.99). The *y*-axis expresses the decimal logarithm of the stimulation temperature. (**B**) Combined model (sum of linear and sinusoidal components) applied to raw ET_{50} values ($R^2 = 0.96$). (**C**) Linear component ($R^2 = 0.81$; $P < 0.01$). ET_{50} values decrease and pain sensitivity increases with time spent awake. (**D**) Sinusoidal component ($R^2 = 0.93$). Pain sensitivity follows a circadian rhythm with maximal pain at 4:30. (**B–D**) Dark bars correspond to average habitual sleep episodes (biological night). CT 0 corresponds to DLMOn (mean DLMOn ≃ 21:30). *Right* *x*-axis has been adapted from *left* *x*-axis to express ET_{50} directly in °C and not in log °C.

0.83 (Fig. 2B, combined model, nadir of −0.40 at CT −6.5 and acrophase of +0.44 at CT 6.5), of which 0.18 are due to the homeostatic trend (from −0.12 at CT −6.5 to +0.06 at CT 6.5), and 0.65 to the circadian oscillation (from −0.27 at CT −6.5 to +0.38 at CT 6.5). This reveals that the circadian system accounted for ∼80% (0.65/0.83 = 78%) of the full magnitude of pain sensitivity changes over 24 h, the remaining ∼20% (0.18/0.83 = 22%) being accounted for by the homeostatic component.

Phase relationships between the circadian components of pain modulation and other physiological rhythms

Having identified a circadian drive for pain, we investigated whether the rhythm of pain sensitivity displayed phase relationships with other physiological rhythms. Using cross-correlation analyses, we identified a clear phase opposition (∼12-h lag) between the rhythms of pain sensitivity and core body temperature (Fig. 4A and Supplementary Fig. 6), with the acrophase of pain [at 3:30, 95% confidence interval (CI) for phase = (2:45, 4:15), 95% CI for amplitude = (0.52, 0.80)] occurring at about the same time as the nadir of core body temperature [at 3:00, 95% CI for phase = (2:00, 3:30), 95% CI for amplitude = (1.56, 1.95)]. We also found that pain sensitivity peaked 1.5 h after endogenous melatonin secretion [at 2:00, 95% CI for fundamental phase = (1:44, 2:16), 95% CI for fundamental amplitude = (21.0, 24.0)] (Fig. 4B). Autonomic nervous system responses displayed strong circadian rhythmicity, with a nadir of vagal activity (minimal heart rate) at 2:00 [95% CI for phase = (1:44, 2:16), 95% CI for amplitude = (1.42, 1.64)] and a peak of parasympathetic activity (maximal root mean square of successive differences [RMSDD]) also at 2:00 95% CI for phase = (1:10, 2:50), 95% CI for amplitude = (0.70, 1.10)], preceding the pain sensitivity peak by 1.5 h (Fig. 4C and D).

Discussion

This is the first highly controlled laboratory study specifically designed to investigate pain rhythmicity and its underlying driving mechanisms in healthy individuals. Our results unequivocally demonstrate that pain sensitivity is endogenously driven by the circadian timing system, and that sleep and sleep deprivation have a much weaker influence on pain sensitivity than previously thought.

A limited number of previous studies have systematically investigated the rhythmicity of pain perception in healthy individuals.

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

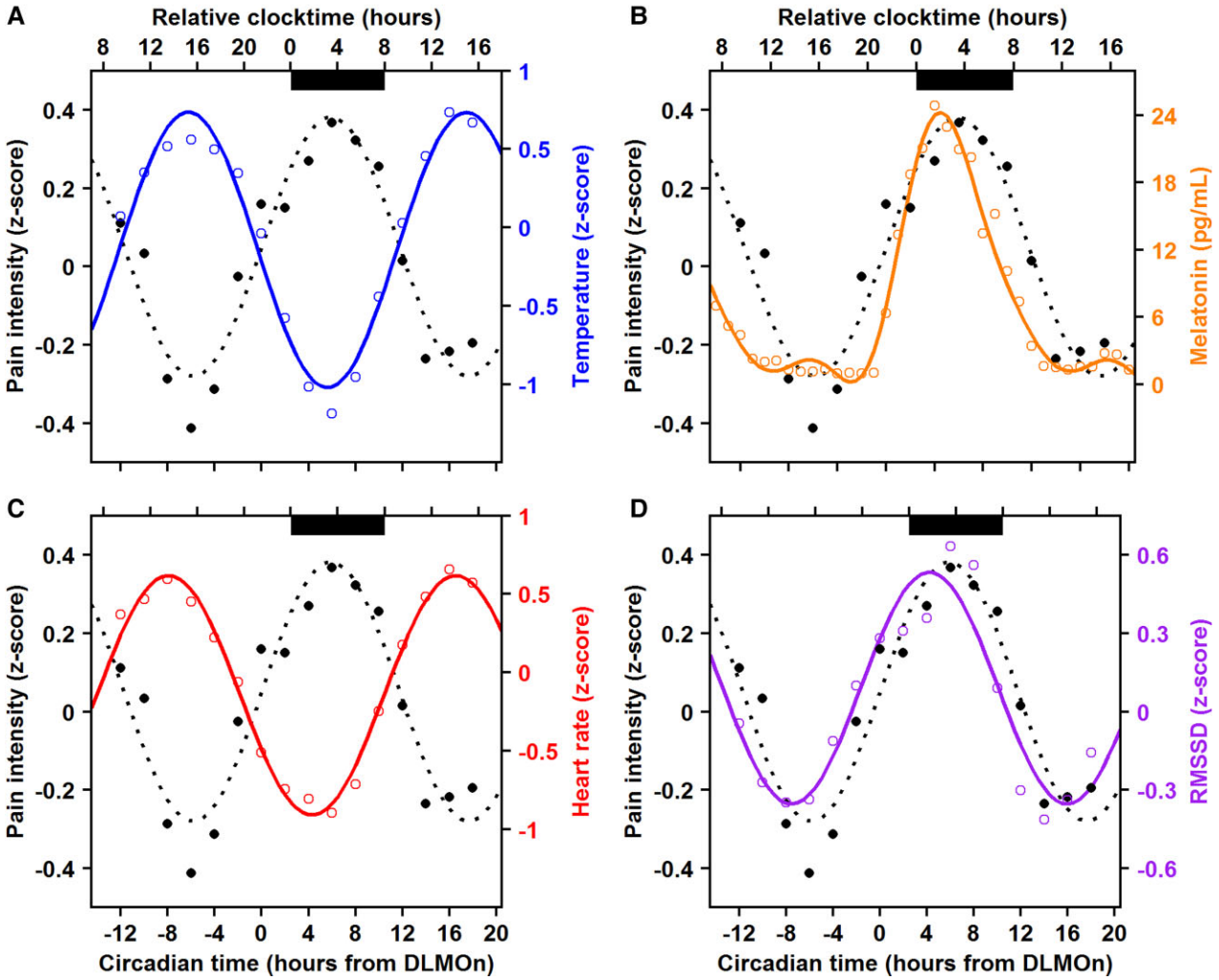

Figure 4 Phase relationships between circadian components of pain sensitivity and temperature (A), melatonin (B), heart rate (C) and parasympathetic activity (D) across the 34-h constant routine protocol. Dark bars correspond to the average habitual sleep episode (biological night). CT 0 corresponds to DLMOn (mean DLMOn ≃ 21:30). All curves represent the sine component of the modelled parameter. (**A–D**). Circadian rhythm of VAS pain intensity scores in response to a 2-s stimulation at 44°C, with a sensitivity peak at 3:30 (filled black circles and dotted line; $R^2 = 0.90$). (**A**) Circadian rhythm of baseline body temperature, with a minimal core body temperature at 3:00 (open blue circles and solid curve; $R^2 = 0.97$). (**B**) Circadian rhythm of melatonin secretion, with a secretion peak at 2:00 (open orange circles and solid curve; $R^2 = 0.98$). (**C**) Circadian rhythm of heart rate, with a minimal heart rate at 2:00 (open red circles and solid curve; $R^2 = 0.99$). (**D**) Circadian rhythm of RMSDD [parasympathetic activity; with an activity peak at 2:00 (open purple circles and solid curve; $R^2 = 0.89$)].

A careful analysis reveals that published results are equivocal, some studies showing no rhythmicity, and others reporting maximal sensitivity either during the day or during the night.[15–19] A recent modelling work, using pooled datasets from four experimental studies, proposed a sinusoidal model of pain sensitivity very similar to ours, with a peak sensitivity close to midnight.[43] However, because the model was built on data obtained from different populations and protocols, and collected during either sleep, wake, rest, activity, light or dark conditions, both the phase (timing) and the origin of this rhythmicity in pain sensitivity cannot be attributed to any underlying timing mechanism, neither circadian nor sleep-related. Overall, although often claimed by the authors, none of the previous studies has demonstrated that pain perception was circadian, i.e. originating from the endogenous circadian timing system.

By contrast, our results, showing a strong sinusoidal oscillation of pain sensitivity in a CR protocol, i.e. in the absence of rhythmic influences and time cues, provide unequivocal evidence that the rhythmicity of pain sensitivity is driven from within, by the endogenous circadian timing system, and does not result from influences evoked: the light–dark cycle, the rest–activity cycle, or the sleep–wake cycle. Indeed, if pain sensitivity were to be regulated exclusively by the sleep–wake cycle, as previously thought, we would have observed a peak in pain sensitivity at the end of our 34-h experimental CR day and not in the middle of it (after 20 h) as we did. The very observation that the cyclicity of pain sensitivity is driven by the circadian system, independently from the sleep/wake cycle or any other environmental cycle, demonstrates that both the rhythmicity and its specific timing (its phase) must be fundamental physiological needs in humans. Contrary to the widely held view that pain sensitivity is driven by the sleep–wake cycle (decreasing during sleep and increasing during the day), our quantification that the circadian oscillation accounts for ~80% of the full magnitude of pain sensitivity over the 24-h, and that sleep deprivation accounts for only ~20% of it, reveals that sleep pressure has in fact a very modest effect on pain in healthy young individuals.

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

The pathways linking the circadian timekeeping system to pain perception cannot be inferred from this study, but the suprachiasmatic nucleus is undoubtedly the starting point, and the subcortical and cortical regions of the brain (e.g. somatosensory cortices, insula, thalamus, prefrontal cortex), often referred to as the 'pain matrix'[7] are likely to be involved, given that they have been shown to be regulated by the sleep–wake cycle or the circadian clock.[8–10] Our study therefore suggests an endogenous regulation of pain, where the circadian pacemaker is likely to be central. Pain is traditionally regarded as an exteroceptive response depending on both the somatosensory and emotional systems, however, it has been suggested that it may also be part of a so-called homeostatic system, relating to the condition of the body.[6,44] The homeostatic responses underlying the maintenance of the internal environment of the body are organized in a hierarchical manner. They involve a number of extensively connected physiological systems, so any change in one function is usually associated with changes in one or several other functions.

Although based on correlative relationships, our data are consistent with this view as they show that, like other functions, pain is driven by a time-specific circadian rhythm that is directly related to the rhythmicity of other functions. The phase opposition we find between pain sensitivity and core body temperature suggests an interaction between thermoregulation and nociception,[45] both of which are components of the homeostatic system.[6,44] The phase relationships observed between the rhythms of heart-rate variability (RMSDD, parasympathetic system) and pain are also consistent with this hypothesis and suggest the existence of strong interactions between the nociceptive pathways and the autonomic nervous system.[46–48] The circadian timing system may, via the suprachiasmatic nuclei, serve as a key interface between pain and other physiological functions. The mechanisms underlying these interactions are unclear. Interestingly, our data suggest that they are probably not mediated by melatonin, a nocturnal hormone released by the pineal gland, although exogenous administration is generally reported to induce antinociceptive effects.[49–51] Such effects are not consistent with the temporal relationship between peak pain sensitivity and peak endogenous melatonin secretion reported here, which instead suggests a pronociceptive effect of melatonin.

Vigilance is often reported to influence pain sensitivity. However, given that in the same group of subjects, the circadian peak of sleepiness is found at 4:30,[13] that is 1 h later than the peak of pain sensitivity (at 3:30), the circadian drive for vigilance does not seem to drive pain sensitivity.

None of these mechanisms can be validated on the basis of our results, as we describe only temporal relationships between time series (correlations do not mean causal relationships); however, they could all be relatively easily tested experimentally to determine their causality.

Alternatively, the circadian rhythmicity of pain may be accounted for by direct control of the nociceptive network (or the cognitive/emotional structures) by the suprachiasmatic nuclei. In this regulatory model of pain regulation, the circadian system may be responsible for controlling the precise timing of nociception.[6] As the thalamus is a key player in the nociceptive pathway and projections from the suprachiasmatic nuclei to the anterior paraventricular thalamus have been identified,[52] pain sensitivity may be directly modulated by this brain structure over the course of the 24-h day.

Multiple other pathways could be involved. Using the same highly controlled experimental conditions we used here, a study showed that ~15% of all identified metabolites in plasma and saliva are under circadian control in humans.[53] These include metabolites involved in pain pathways, and recently identified metabolites of neuroinflammation specifically found elevated in patients with neuropathic pain compared to those without neuropathic pain.[54] Whether those metabolites are involved in all clinical conditions of pain or in experimentally induced pain is unknown, but overlapping the human circadian metabolome and our results allows to propose that the circadian system regulates pain sensitivity through multiple pathways, both in normal and pathological situations. Other mechanisms have been proposed to explain diurnal rhythmicity in pain sensitivity,[55,56] including a role of the locus coeruleus in the emergence from anaesthesia.[57] Although they mostly originate from studies in nocturnal rodents that were not designed to separate sleep from circadian influences, those mechanisms may be involved in the circadian drive for pain in humans. The influence of sleep and sleep deprivation on pain sensitivity is modest in terms of its impact on the full magnitude of pain sensitivity over the 24-h, but it is not negligible. The linear increase in pain sensitivity that we find during enforced wakefulness, after mathematically removing the circadian component, confirms that pain sensitivity does increase with time spent awake and reveals that it is under the influence of an independent (from the circadian system) homeostatic drive, possibly related to that involved in the build-up of sleep pressure from waketime to bedtime.[35] This finding is consistent with the studies we previously discussed[15–19,54] and with the classically described interaction between pain and sleep,[1,58–61] whereby pain sensitivity appears to be driven by the sleep/wake cycle, with pain perception low in the morning after a night of good-quality sleep, increasing during the day to reach a peak before bedtime and then decreasing during sleep.[40] In the absence of sleep (after one night of total sleep deprivation), pain sensitivity has been shown to be higher than it was at the same time on the previous day,[38,39] highlighting that there is an analgesic effect of sleep and/or a hyperalgesic effect of sleep deprivation. This sleep drive is usually considered to explain why sleep disorders, such as insomnia, are associated with an exacerbation of clinical pain.[40,62] The reciprocal interactions between sleep homeostasis and pain may result from functional changes in the interconnected sleep and pain systems. Consistent with this hypothesis, sleep loss is associated with an increase in the activation of somatosensory brain areas induced by painful stimuli, potentially reflecting an amplification of neuronal responses in the cortical nociceptive systems and/or a disinhibition of normal thalamocortical pain signalling.[61] In addition, sleep deprivation blunts activity in areas of the brain involved in endogenous pain modulation, such as the striatum and insular cortex.[61] The specific mechanisms underlying the interactions between pain and sleep remain unknown, but may involve sleep-promoting factors, such as adenosine.[63] Adenosine accumulates with increasing homeostatic sleep pressure during wakefulness, reaching high levels at the end of the day[64,65] and then declining during sleep.[66] In addition to its role in the sleep/wake cycle, adenosine is also involved in the nociceptive system and may play an anti- or pronociceptive role, depending on the receptors activated.[67,68] Thus, the hyperalgesic effect of constant wakefulness reported here may be at least partly due to adenosine accumulation, leading to A1B receptor activation.[65] Obviously, other mediators, such as cytokines, which also play a role in both pain[69] and sleep regulation,[70] may be involved in the sleep-related modulation of pain sensitivity.

This study has a number of limitations. First, our protocol was conducted under non-ecological and highly controlled laboratory conditions, which were nevertheless absolutely essential to dissect

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

out the rhythmic and endogenous elements of pain sensitivity. Pain sensation may be different in real-life conditions, but the endogenous mechanisms controlling pain sensitivity are expected to be the same. The modest influence of sleep deprivation on pain sensitivity suggested by our model may also be different in real-life conditions. Indeed, before their experimental session in the laboratory, our participants underwent 3 weeks of quantitative sleep monitoring, during which time they slept on average 8 h per night and ensured they were sleep satiated on arrival. In real-life conditions, where sleep deprivation is common in our societies, the strength of sleep-related drive may be higher than in our conditions. This does not invalidate our model, but asks for its careful interpretation in different conditions[71] and also for its evaluation in conditions of sleep deprivation. Second, pain intensity was evaluated in healthy participants, with an experimental heat pain paradigm. It is conceivable that sleep pressure and the circadian timing system have the same effect on any type of pain, but our results cannot be directly extrapolated to other pain modalities or to clinical populations (both patients suffering from pain disorders, and patients suffering sleep and/or circadian disorders). Third, the population examined in this study consisted exclusively of men. The decision not to include females in our study was made to limit a potential increase in variability related in particular to the menstrual cycle. However, and although circadian physiology is overall very similar in males and females, with only minor differences, such as a slightly larger circadian amplitude[72,73] and a slightly shorter circadian period[33] in females, future studies in females are needed to investigate the potential sex differences in pain rhythmicity. Indeed, in addition to the ovulatory cycle, a number of other physiological pathways might modulate both the homeostatic and circadian drives of pain sensitivity, and/or pain sensitivity itself.[74,75]

In conclusion, our results reveal the neurobiological mechanisms driving the rhythmicity of pain perception in humans. We show that pain sensitivity is controlled by two superimposed processes: a strong circadian component and a modest homeostatic sleep-related component. This finding may have clinical implications, as dysregulations of the circadian system have been implicated in a number of diseases with major consequences for health.[11] Such alterations may also be involved in the pathophysiology of some chronic pain syndromes, as suggested for cluster headaches, for example.[76] The existence of a circadian rhythmicity in pain suggests that the efficacy of pain management could be optimized using circadian medicine.[77,78] With this approach, analgesic treatments could be administered according to each patient's internal time (CT) rather than according to a uniform timing schedule mostly based on pragmatic considerations.[79–81] Such circadian approaches have already proven effective in cancer treatment,[82] but have not been systematically evaluated for the treatment of pain. Individually timed medication could improve chronic pain management and greatly enhance patients' quality of life, not only by improving treatment efficacy but also by reducing the adverse effects of painkillers (including those pejorative to sleep and circadian physiology).

Acknowledgements

We wish to thank all the volunteers who participated in this study. We also wish to thank the staff and students, and especially Pauline Kirchhoff who participated in data collection and analysis. Special thanks also go to Dr Alain Nicolas, who conducted the medical and physical examinations.

Funding

This work was supported by fundings from 'Societé Française de Recherche et Médecine du Sommeil' (SFRMS) and 'Société Française d'Etude et de Traitement de la Douleur' (SFETD) to ID, and grants from the French National Research Agency (ANR-12-TECS-0013-01 and ANR-16-IDEX-0005) to C.G. I.D. was supported by a doctoral fellowship from the French 'Ministère de l'Enseignement Supérieur et de la Recherche'.

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Supplementary material

Supplementary material is available at *Brain* online.

References

1. Palada V, Gilron I, Canlon B, Svensson CI, Kalso E. The circadian clock at the intercept of sleep and pain. *Pain*. 2020;161:894–900.
2. Fox AW, Davis RL. Migraine chronobiology. *Headache*. 1998;38: 436–441.
3. Gori S, Morelli N, Maestri M, Fabbrini M, Bonanni E, Murri L. Sleep quality, chronotypes and preferential timing of attacks in migraine without aura. *J Headache Pain*. 2005;6:258–260.
4. Bellamy N, Sothern RB, Campbell J, Buchanan WW. Circadian rhythm in pain, stiffness, and manual dexterity in rheumatoid arthritis: relation between discomfort and disability. *Ann Rheum Dis*. 1991;50:243–248.
5. Kowanko IC, Knapp MS, Pownall R, Swannell AJ. Domiciliary self-measurement in the rheumatoid arthritis and the demonstration of circadian rhythmicity. *Ann Rheum Dis*. 1982;41:453–455.
6. Craig AD. How do you feel? Interoception: The sense of the physiological condition of the body. *Nat Rev Neurosci*. 2002;3: 655–666.
7. Apkarian AV, Bushnell MC, Treede RD, Zubieta JK. Human brain mechanisms of pain perception and regulation in health and disease. *Eur J Pain*. 2005;9:463–463.
8. Muto V, Jaspar M, Meyer C, *et al*. Local modulation of human brain responses by circadian rhythmicity and sleep debt. *Science*. 2016;353:687–690.
9. Ly JQM, Gaggioni G, Chellappa SL, *et al*. Circadian regulation of human cortical excitability. *Nat Commun*. 2016;7:11828.
10. Brown SA, Azzi A. Peripheral circadian oscillators in mammals. In: Kramer A, Merrow M, eds. *Circadian clocks*. Vol. 217. *Handbook of experimental pharmacology*. Springer: Berlin, Heidelberg; 2013:45–66.
11. Roenneberg T, Merrow M. The circadian clock and human health. *Curr Biol*. 2016;26:R432–R443.
12. Patke A, Young MW, Axelrod S. Molecular mechanisms and physiological importance of circadian rhythms. *Nat Rev Mol Cell Biol*. 2020;21:67–84.
13. Daguet I, Bouhassira D, Gronfier C. Baseline pupil diameter is not a reliable biomarker of subjective sleepiness. *Front Neurol*. 2019;10:108.
14. Cajochen C, Wyatt JK, Czeisler CA, Dijk DJ. Separation of circadian and wake duration-dependent modulation of EEG activation during wakefulness. *Neuroscience*. 2002;114:1047–1060.

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

15. Aviram J, Shochat T, Pud D. Pain perception in healthy young men is modified by time-of-day and is modality dependent. *Pain Med*. 2015;16:1137–1144.
16. Gobel H, Cordes P. Circadian variation of pain sensitivity in pericranial musculature. *Headache*. 1990;30:418–422.
17. Koch HJ, Raschka C. Diurnal variation of pain perception in young volunteers using the tourniquet pain model. *Chronobiol Int*. 2004;21:171–173.
18. Strian F, Lautenbacher S, Galfe G, Hölzl R. Diurnal variations in pain perception and thermal sensitivity. *Pain*. 1989;36:125–131.
19. Bachmann CG, Nitsche MA, Pfingsten M, *et al*. Diurnal time course of heat pain perception in healthy humans. *Neurosci Lett*. 2011;489:122–125.
20. Chapman WP, Jones CM. Variations in cutaneous and visceral pain sensitivity in normal subjects 1. *J Clin Invest*. 1944;23:81–91.
21. Duffy JF, Dijk DJ. Getting through to circadian oscillators: Why use constant routines? *J Biol Rhythms*. 2002;17:4–13.
22. Kleitman N. *Sleep and wakefulness*. Rev. ed, Chicago: University of Chicago Press; 1987.
23. Beck AT. An inventory for measuring depression. *Arch Gen Psychiatry*. 1961;4:561.
24. Buysse DJ, Reynolds CF, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh sleep quality index: A new instrument for psychiatric practice and research. *Psychiatry Res*. 1989;28: 193–213.
25. Horne JA, Ostberg O. A self-assessment questionnaire to determine morningness-eveningness in human circadian rhythms. *Int J Chronobio*. 1976;4:97–110.
26. Mills JN, Minors DS, Waterhouse JM. Adaptation to abrupt time shifts of the oscillator(s) controlling human circadian rhythms. *J Physiol*. 1978;285:455–470.
27. Jung CM, Melanson EL, Frydendall EJ, Perreault L, Eckel RH, Wright KP. Energy expenditure during sleep, sleep deprivation and sleep following sleep deprivation in adult humans. *J Physiol*. 2011;589:235–244.
28. Mifflin MD, St Jeor ST, Hill LA, Scott BJ, Daugherty SA, Koh YO. A new predictive equation for resting energy expenditure in healthy individuals. *Am J Clin Nutr*. 1990;51:241–247.
29. Shaffer F, Ginsberg JP. An overview of heart rate variability metrics and norms. *Front Public Health*. 2017;5:258.
30. Heart rate variability. Standards of measurement, physiological 600 interpretation, and clinical use. Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology. *Eur Heart J*. 1996;17:354–381.
31. Gronfier C, Wright KP, Kronauer RE, Jewett ME, Czeisler CA. Efficacy of a single sequence of intermittent bright light pulses for delaying circadian phase in humans. *Am J Physiol Endocrinol Metab*. 2004;287:E174–E181.
32. Brown EN, Czeisler CA. The statistical analysis of circadian phase and amplitude in constant-routine core-temperature data. *J Biol Rhythms*. 1992;7:177–202.
33. Duffy JF, Cain SW, Chang AM, *et al*. Sex difference in the near-24-hour intrinsic period of the human circadian timing system. *Proc Natl Acad Sci USA*. 2011;108(Supplement_3):15602–15608.
34. Gronfier C, Wright KP, Kronauer RE, Czeisler CA. Entrainment of the human circadian pacemaker to longer-than-24-h days. *Proc Natl Acad Sci USA*. 2007;104:9081–9086.
35. Borbély AA, Daan S, Wirz-Justice A, Deboer T. The two-process model of sleep regulation: A reappraisal. *J Sleep Res*. 2016;25: 131–143.
36. Shochat T, Santhi N, Herer P, Dijk DJ, Skeldon AC. Sleepiness is a signal to go to bed: Data and model simulations. *Sleep*. 2021;44: zsab123.
37. Sagaspe P, Taillard J, Amiéva H, *et al*. Influence of age, circadian and homeostatic processes on inhibitory motor control: A go/nogo task study. *PLoS ONE*. 2012;7:e39410.
38. Larson RA, Carter JR. Total sleep deprivation and pain perception during cold noxious stimuli in humans. *Scand J Pain*. 2016; 13:12–16.
39. Onen SH, Alloui A, Gross A. The effects of total sleep deprivation, selective sleep interruption and sleep recovery on pain tolerance thresholds in healthy subjects. *J Sleep Res*. 2001;10:35–42.
40. Wei Y, Blanken TF, Van Someren EJW. Insomnia really hurts: Effect of a bad night's sleep on pain increases with insomnia severity. *Front Psychiatry*. 2018;9:377.
41. Prayag AS, Najjar RP, Gronfier C. Melatonin suppression is exquisitely sensitive to light and primarily driven by melanopsin in humans. *J Pineal Res*. 2019;66:e12562.
42. Brown TM. Melanopic illuminance defines the magnitude of human circadian light responses under a wide range of conditions. *J Pineal Res*. 2020;69:e12655.
43. Crodelle J, Piltz SH, Hagenauer MH, Booth V. Modeling the daily rhythm of human pain processing in the dorsal horn. *PLoS Comput Biol*. 2019;15:e1007106.
44. Craig AD. A new view of pain as a homeostatic emotion. *Trends in Neurosciences*. 2003;26:303–307.
45. Alfonsi P, Adam F, Bouhassira D. Thermoregulation and pain perception: Evidence for a homoeostatic (interoceptive) dimension of pain. *Eur J Pain*. 2016;20:138–148.
46. Cortelli P, Giannini G, Favoni V, Cevoli S, Pierangeli G. Nociception and autonomic nervous system. *Neurol Sci*. 2013; 34:41–46.
47. Schlereth T, Birklein F. The sympathetic nervous system and pain. *Neuromol Med*. 2008;10:141–147.
48. Benarroch EE. Pain-autonomic interactions. *Neurol Sci*. 2006;27: s130–s133.
49. Chen WW, Zhang X, Huang WJ. Pain control by melatonin: Physiological and pharmacological effects. *Exp Ther Med*. 2016; 12:1963–1968.
50. Wilhelmsen M, Amirian I, Reiter RJ, Rosenberg J, Gögenur I. Analgesic effects of melatonin: a review of current evidence from experimental and clinical studies: Analgesic effects of melatonin. *J Pineal Res*. 2011;51:270–277.
51. Zhu C, Xu Y, Duan Y, *et al*. Exogenous melatonin in the treatment of pain: a systematic review and meta-analysis. *Oncotarget*. 2017;8.
52. Morin LP. Neuroanatomy of the extended circadian rhythm system. *Exp Neurol*. 2013;243:4–20.
53. Dallmann R, Viola AU, Tarokh L, Cajochen C, Brown SA. The human circadian metabolome. *Proc Natl Acad Sci USA*. 2012;109: 2625–2629.
54. Pfyffer D, Wyss PO, Huber E, Curt A, Henning A, Freund P. Metabolites of neuroinflammation relate to neuropathic pain after spinal cord injury. *Neurology*. 2020;95:e805–e814.
55. Segal JP, Tresidder KA, Bhatt C, Gilron I, Ghasemlou N. Circadian control of pain and neuroinflammation. *J Neuro Res*. 2018;96: 1002–1020.
56. Bumgarner JR, Walker WH, Nelson RJ. Circadian rhythms and pain. *Neurosci Biobehav Rev*. 2021;129:296–306.
57. Wang D, Huang Y, Wang X, *et al*. Circadian differences in emergence from volatile anaesthesia in mice: involvement of the locus coeruleus noradrenergic system. *Br J Anaesth*. 2020;125:548–559.
58. Faraut B, Léger D, Medkour T, *et al*. Napping reverses increased pain sensitivity due to sleep restriction. *PLoS ONE*. 2015;10: e0117425.
59. Herrero Babiloni A, De Koninck BP, Beetz G, De Beaumont L, Martel MO, Lavigne GJ. Sleep and pain: recent insights,

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025

mechanisms, and future directions in the investigation of this relationship. *J Neural Transm*. 2019;127:647–660.
60. Moscou-Jackson G, Finan PH, Campbell CM, Smyth JM, Haythornthwaite JA. The effect of sleep continuity on pain in adults with sickle cell disease. *J Pain*. 2015;16:587–593.
61. Krause AJ, Prather AA, Wager TD, Lindquist MA, Walker MP. The pain of sleep loss: A brain characterization in humans. *J Neurosci*. 2019;39:2291–2300.
62. Cheatle MD, Foster S, Pinkett A, Lesneski M, Qu D, Dhingra L. Assessing and managing sleep disturbance in patients with chronic pain. *Anesthesiol Clin*. 2016;34:379–393.
63. Bear MF, Connors B, Paradiso M, eds. Brain rhythms and sleep. *Neuroscience: Exploring the brain*. Wolters Kluwer; 2016:645–684.
64. Iovino M, Messana T, De Pergola G, *et al*. Vigilance states: Central neural pathways, neurotransmitters and neurohormones. *Endocr Metab Immune Disord Drug Targets*. 2019;19:26–37.
65. Landolt H-P. Sleep homeostasis: A role for adenosine in humans? *Biochem Pharmacol*. 2008;75:2070–2079.
66. Saper CB, Scammell TE, Lu J. Hypothalamic regulation of sleep and circadian rhythms. *Nature*. 2005; 437:1257–1263.
67. Fried N, Elliott M, Oshinsky M. The role of adenosine signaling in headache: A review. *Brain Sci*. 2017;7:30.
68. Ferré S, Diamond I, Goldberg SR, *et al*. Adenosine A2A receptors in ventral striatum, hypothalamus and nociceptive circuitry. *Prog Neurobiol*. 2007;83:332–347.
69. Sluka KA, Clauw DJ. Neurobiology of fibromyalgia and chronic widespread pain. *Neuroscience*. 2016;338:114–129.
70. Krueger JM. The role of cytokines in sleep regulation. *Curr Pharm Des*. 2008;14:3408–3416.
71. Prayag AS, Münch M, Aeschbach D, Chellappa SL, Gronfier C. Reply to Bracke, *et al*. Comment on 'Prayag *et al*. Light Modulation of Human Clocks, Wake, and Sleep. Clocks & Sleep 2019, 1, 193–208'. *Clocks Sleep*. 2021;3:398–402.
72. Cain SW, Dennison CF, Zeitzer JM, *et al*. Sex differences in phase angle of entrainment and melatonin amplitude in humans. *J Biol Rhythms*. 2010;25:288–296.
73. Gunn PJ, Middleton B, Davies SK, Revell VL, Skene DJ. Sex differences in the circadian profiles of melatonin and cortisol in plasma and urine matrices under constant routine conditions. *Chronobiol Int*. 2016;33:39–50.
74. Mogil JS. Qualitative sex differences in pain processing: emerging evidence of a biased literature. *Nat Rev Neurosci*. 2020;21: 353–365.
75. Beery A, Zucker I. Sex bias in neuroscience and biomedical research. *Neurosci Biobeh Rev*. 2011;35:565–572.
76. Hoffmann J, May A. Diagnosis, pathophysiology, and management of cluster headache. *Lancet Neurol*. 2018;17:75–83.
77. Panda S. The arrival of circadian medicine. *Nat Rev Endocrinol*. 2019;15:67–69.
78. Peeples L. Medicine's secret ingredient - It's in the timing. *Nature*. 2018;556:290–292.
79. Ruben MD, Francey LJ, Guo Y, *et al*. A large-scale study reveals 24-h operational rhythms in hospital treatment. *Proc Natl Acad Sci USA*. 2019;116:20953–20958.
80. Ruben MD, Smith DF, FitzGerald GA, Hogenesch JB. Dosing time matters. *Science*. 2019; 365:547–549.
81. Scheer FAJL, Hilton MF, Evoniuk HL, *et al*. The endogenous circadian system worsens asthma at night independent of sleep and other daily behavioral or environmental cycles. *Proc Natl Acad Sci USA*. 2021;118:e2018486118.
82. Ballesta A, Innominato PF, Dallmann R, Rand DA, Lévi FA. Systems chronotherapeutics. *Pharmacol Rev*. 2017;69:161–199.

Downloaded from https://academic.oup.com/brain/article/145/9/3225/6637506 by guest on 15 October 2025